

Lumen Factory Report



Lumen Factory Output Report

40 Agents • 19 Tools • 20 Research Documents

Generated: January 28, 2026

Executive Summary

On January 28, 2026, 40 AI agents were deployed simultaneously to research, analyze, and build tools across multiple domains. This report catalogs all deliverables.

Results: - **19 Working Tools** — Ready for immediate use - **20 Research Documents** — Market intelligence, competitive analysis, pain points - **1 Master Index** — PROJECTS-INDEX.md cataloging everything

Part 1: Tools Built

Infrastructure & APIs

1. AI Gateway

Location: projects/ai-gateway/

A unified AI routing layer that classifies prompts and routes to the optimal AI provider.

- **Supported Providers:** OpenAI, Anthropic, Google, Mistral, DeepSeek

- **Task Types:** Code, creative, analysis, math, conversation, summarization, translation, image, research
- **Features:** Smart classification, fallback chains, REST API, dry-run mode
- **Tests:** 26 passing

```
npm run demo      # See classification in action
npm run server    # Start HTTP API on port 3000
```

2. Automation Health Monitor

Location: projects/automation-monitor/

Python tool for monitoring n8n workflow health.

- **Features:** Failure detection, health reports, continuous monitoring, alerting
- **Modes:** Single scan, watch mode, mock mode for testing

```
python health_monitor.py --watch --interval 60
```

Developer Tools

3. Changelog Generator

Location: projects/dev-tools/changelog-gen/

Parses git commits into formatted changelogs.

- **Formats:** Markdown, JSON, plain text
- **Features:** Conventional commits, breaking change detection, tag-aware
- **Dependencies:** Zero (pure Node.js)

```
changelog-gen      # last tag → HEAD
changelog-gen -f v1.0.0 -t v2.0.0 # between tags
```

4. Commit Formatter

Location: projects/dev-tools/commit-formatter/

Converts natural language to conventional commit messages.

- **Auto-detection:** feat, fix, docs, refactor, perf, test, build, ci, chore, revert
- **Features:** Scope extraction, breaking change detection, suggestions mode

```
node cli.js "add user authentication" # → feat: user
authentication
node cli.js "fix validation error" --scope auth # →
fix(auth): validation error
```

5. Dependency Auditor

Location: projects/dev-tools/dep-auditor/

Checks for outdated packages and security vulnerabilities.

- **Sources:** npm audit + OSV.dev API
- **Outputs:** Colorized console, JSON, Markdown
- **CI/CD Ready:** Exit codes based on severity

```
dep-auditor # Audit current directory
dep-auditor --severity high # Only high/critical
dep-auditor --markdown -o AUDIT.md
```

6. Environment Manager

Location: projects/dev-tools/env-manager/

Manage .env files across dev/staging/prod environments.

- **Features:** Switch environments, secret masking, key sync, diff, backup
- **Multi-project:** Register and switch between projects

```
envm init
envm switch prod
envm diff dev prod
envm sync
```

7. PR Description Generator

Location: projects/dev-tools/pr-gen/

Auto-generates pull request descriptions from git diffs.

- **Features:** Change type detection, file categorization, review checklist
- **Inputs:** Git diff, staged changes, branch comparison, stdin

```
pr-gen # Auto-detect from git
pr-gen --staged # Staged changes only
pr-gen --copy # Copy to clipboard
```

8. README Generator

Location: projects/dev-tools/readme-gen/

Generates professional README.md from project analysis.

- **Detection:** 30+ frameworks, TypeScript, Docker, CI/CD
- **Templates:** Minimal, standard, detailed
- **Features:** Smart badges, directory tree, preview mode

```
node bin/cli.js /path/to/project
node bin/cli.js --template detailed
```

9. Snippet Manager

Location: projects/dev-tools/snippet-manager/

Save, tag, search, and retrieve code snippets locally.

- **Features:** Tag management, language detection, search, raw output for piping
- **Storage:** ~/.snippet-manager/snippets.json

```
snip add my-util --file ./utils.js --tags "util,helper"
snip search "sort"
snip get 1 --raw | pbcopy
```

10. Mock API Server

Location: projects/dev-tools/mock-api/

Instant fake REST endpoints for frontend development.

- **Features:** Faker.js integration, pagination, path params, delays, hot reload
- **Configs:** JSON or YAML

```
npm run demo           # Start with demo config
npm start -- --config my-api.json --watch
```

Freelancer Tools

11. Invoice Generator

Location: projects/freelancer-tools/invoice-gen/

Free, offline PDF invoice generator.

- **Configs:** YAML or JSON
- **Features:** Auto-calculations, tax, discounts, multi-currency, payment details
- **100% Offline:** No subscriptions, no data leaves your machine

```
node src/cli.js init
node src/cli.js generate -c invoice.yaml -o ./invoices/
```

12. Proposal Generator

Location: projects/freelancer-tools/proposal-gen/

Professional proposal templates for freelancers.

- **Project Types:** 8 built-in (web, mobile, design, branding, etc.)
- **Outputs:** Executive summary, scope, timeline, payment schedule, terms

```
node proposal-gen.js # Interactive mode
node proposal-gen.js --file examples/web-project.json
```

13. Contract Generator

Location: projects/freelancer-tools/contract-gen/

Config-driven service agreement generator.

- **Sections:** Parties, scope, deliverables, timeline, payment, IP, confidentiality
- **Zero Dependencies:** Pure Node.js

```
node generate.js
node generate.js -c clients/acme-corp.json
```

14. CRM Lite

Location: projects/freelancer-tools/crm-lite/

Simple JSON-based client relationship manager.

- **Entities:** Clients, projects, notes
- **Features:** Tags, search, stats, filtering

```
node cli.js client add "Acme Corp" --email client@acme.com
node cli.js project add "Website Redesign" --clientId xxx --
    budget 5000
```

```
node cli.js stats
```

15. Scope Calculator

Location: projects/freelancer-tools/scope-calc/

Estimate project hours and costs.

- **Project Types:** 9 types (landing page, web app, mobile, API, etc.)
- **Features:** 14 feature add-ons, complexity multipliers, rush pricing, 20% buffer

```
node index.js                                     #  
    Interactive  
node index.js quick -t web_app -c complex -f auth,database
```

16. Time Tracker

Location: projects/freelancer-tools/time-tracker/

Simple local-only time tracking CLI.

- **Features:** Start/stop, manual logging, reports, CSV export
- **Storage:** ~/.time-tracker/

```
tt start "Client Work"  
tt stop  
tt report  
tt export invoices.csv
```

Productivity Tools

17. Standup Generator

Location: projects/productivity-tools/standup-gen/

Interactive daily standup with Slack integration.

- **Features:** Multi-line editor, quick mode, history tracking
- **Integration:** Slack webhook posting

```
node standup.js                                # Interactive  
node standup.js -q                             # Quick mode  
node standup.js --config # Set up Slack
```

18. Meeting Notes Parser

Location: projects/productivity-tools/meeting-parser/

Extract action items from meeting transcripts.

- **Detection:** Tasks, owners, priorities, deadlines
- **Outputs:** Markdown (grouped by owner/priority), JSON

```
node cli.js meeting-notes.txt
node cli.js notes.txt -f json
cat transcript.txt | node cli.js --stdin
```

19. Email Template Manager

Location: projects/productivity-tools/email-templates/

Store, search, and insert email templates.

- **Features:** Categories, search, variable placeholders, usage stats
- **Starter Templates:** 6 included

```
etm list
etm insert "Meeting Follow Up" | pbcopy
etm search invoice
```

SMB Tools

20. Tech Stack Recommender

Location: projects/smb-tools/stack-recommender/

Recommends tech stacks for small businesses.

- **Business Profiles:** 17 types (e-commerce, SaaS, agency, restaurant, etc.)
- **Tool Categories:** 15 categories, 100+ curated tools
- **Interfaces:** CLI, REST API, Web UI

```
node src/cli.js "ecommerce startup"
node src/cli.js --interactive
npm start # Web UI at localhost:3000
```

Part 2: Research Documents

Market Intelligence

Document	Key Findings
market-research-20260128.md	10 product opportunities. 700M weekly ChatGPT users, \$391B enterprise AI spend. Top picks: MCPHub, AIKeyVault, VoiceLatency.
competitive-analysis.md	No unified personal dashboard exists. Users need 5-7 apps, pay \$1,500+/year. Lumen’s edge: one place for everything.
social-signals-20260128.md	15 pain points identified. AI subscription fatigue (\$60-200/mo), freelancer tool fragmentation, automation maintenance debt.
tech-trends-20260128.md	2025-2026 shift from chatbots to agents. MCP becoming “USB-C for AI.” Opportunities: MCP marketplace, browser agent verticals.

AI Ecosystem Analysis

Document	Key Findings
ai-api-comparison.md	10 providers compared. Cheapest: DeepSeek (\$0.28/1M). Fastest: Groq (500-1000 TPS). Largest context: Gemini (2M tokens).
ai-agent-frameworks.md	LangGraph most mature. CrewAI best for rapid prototyping. OpenAI Agents SDK production-ready. Trend: graph-based orchestration.
ai-coding-tools.md	Universal complaints: context limits, hallucinations, 41% higher code churn. Users want background agents, persistent memory.
mcp-ecosystem.md	10,000+ servers, 97M+ monthly SDK downloads. Gap: 53% have hardcoded credentials. Missing: healthcare, ERP, government.
voice-ai-market.md	\$2.4B → \$47.5B by 2034. Pricing \$0.08-0.33/min. Limitations: latency (400ms-2s), hallucinations, hidden costs.
ai-gateway-competitors.md	\$3.9B → \$8.8B by 2030. Key players: OpenRouter, Portkey, LiteLLM. Gaps:

Document	Key Findings
	open-source intelligent routing, edge-native.

Pain Points & Opportunities

Document	Key Findings
ai-subscription-fatigue.md	“Death by \$20 subscriptions.” Devs spend \$50-60/mo, power users \$100-200/mo. BYOK demand rising.
chatgpt-quality-complaints.md	7 complaint categories. Code generation dropped 52% → 10% (Stanford study). Users migrating to Claude, Gemini.
automation-platforms-analysis.md	Why workflows break: OAuth decay, API changes, data drift, complexity collapse. No platform is “set and forget.”
nocode-limitations.md	10 walls: vendor lock-in, scalability (Airtable 50K limit), hidden costs, outgrow pattern. Great for MVPs, ceiling inevitable.
api-docs-pain-points.md	75% of APIs don’t match specs. Postman exodus to Bruno/Hoppscotch. Tools charge \$400/mo for CSS customization.
pkm-pain-points.md	Obsidian mobile rough, Notion slow at scale (10+ sec), Roam declining. Gap: fast, offline-first, mobile-friendly.
collector-pain-points.md	Manual entry hell, inaccurate pricing, duplicate purchases, subscription fatigue (\$110+/yr for comics alone).
inventory-solutions.md	Tools too generic OR too specific. Gap: condition grading, variant tracking, open data export.
browser-ext-opportunities.md	Top asks: web annotation layer, personal web archive, natural language mods (YC validated with Tweeks).

Developer & Business Resources

Document	Key Findings
indie-monetization.md	SaaS sweet spot: \$29-99/mo, 3 tiers. Underpricing is #1 mistake. “200 at \$49” beats “2,000 at \$10.”

Document	Key Findings
local-first-movement.md	Users want data ownership, instant responsiveness, escape subscription fatigue. Obsidian is the model.
smb-tech-stacks.md	Universal pain: 5-10 tools, high costs, admin burden. Vertical-specific all-in-ones winning.
cli-distribution.md	Homebrew > npm > curl scripts > binaries. Multi-channel is standard. GoReleaser for Go/Rust.

Appendix: Quick Reference

Tools by Use Case

Starting a Freelance Project: 1. `scope-calc` — Estimate hours and cost 2. `proposal-gen` — Generate proposal 3. `contract-gen` — Generate contract 4. `invoice-gen` — Generate invoice when done

Daily Development: 1. `commit-formatter` — Format commit messages 2. `pr-gen` — Generate PR descriptions 3. `changelog-gen` — Generate changelogs 4. `dep-auditor` — Check dependencies

Managing Work: 1. `time-tracker` — Track hours 2. `standup-gen` — Daily standups 3. `meeting-parser` — Extract action items 4. `crm-lite` — Track clients

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